# Table of Contents

## Chapter 12: Reactions of Alkenes

**Oxymercuration-Demercuration**.....2

**Catalytic Hydrogenation**.....3

**Vicinal Dihalogenation**.....4

**Hydrohalogenation**.....5

**Hydroboration - Oxidation**.....6

**Acid-Catalyzed Hydration of Alkenes**.....7

**Haloalcohol (Halohydrin) Formation from Alkenes**.....8

**Haloether Formation from Alkenes**.....9

**Epoxide Synthesis**.....10

**[Epoxide-Hydrolysis-Mediated] *Vicinal Anti-Hydroxylation of Alkenes***.....11

**Vicinal Syn-Dihydroxylation**.....12

**Ozonolysis**.....13

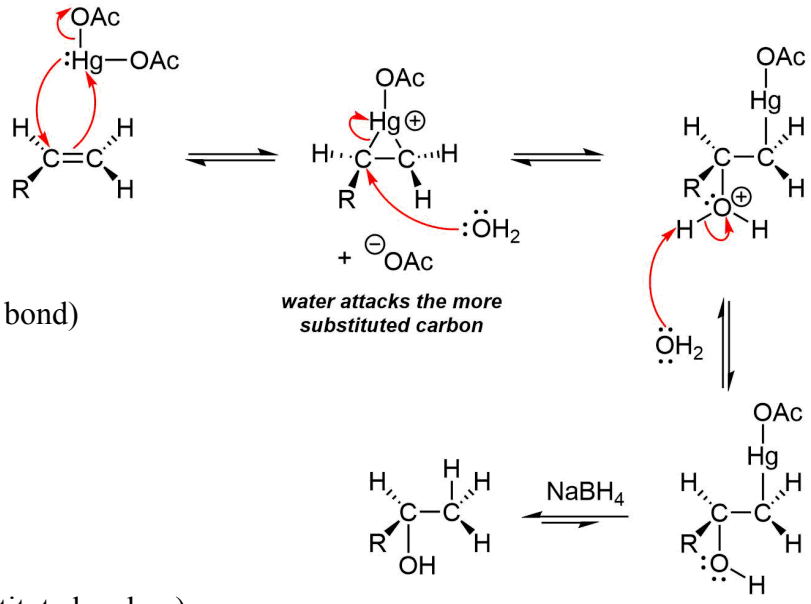
# Oxymercuration-Demercuration→Reduction

**Brief Description:**

The alkene π bond attacks $Hg^{2+}$, forming a bridged mercurinium ion. Water attacks the more substituted carbon (where positive character is greater), opening the ring. Deprotonation gives an alcohol with Hg attached. $NaBH_4$ then **replaces Hg with H**, completing Markovnikov hydration without forming a carbocation.

**Reagent(s):**

1. $Hg(OAc)_2$, $H_2O$
2. $NaBH_4$, NaOH




**Product(s)**

- Alcohol (OH + H across double bond)

**Intermediate(s):**

- Mercurinium ion (bridged)

**Regiochemistry:**

- Markovnikov (OH to more substituted carbon)

**Stereochemistry:**

- Anti (overall)

**Notes:**

- No carbocation → no rearrangements

# Catalytic Hydrogenation

**Brief Description:**

The alkene and $H_2$ both adsorb onto a metal surface. The H–H bond breaks, and both hydrogens are delivered to the same face of the alkene as the π bond is reduced, forming two C–H bonds simultaneously.

**Reagent(s):**

1. $H_2$ + Pd/C (or Pt, Ni)

**Product(s)**

- Alkane (H + H added across DB)

**Intermediate(s):**

- No intermediate

**Regiochemistry:**

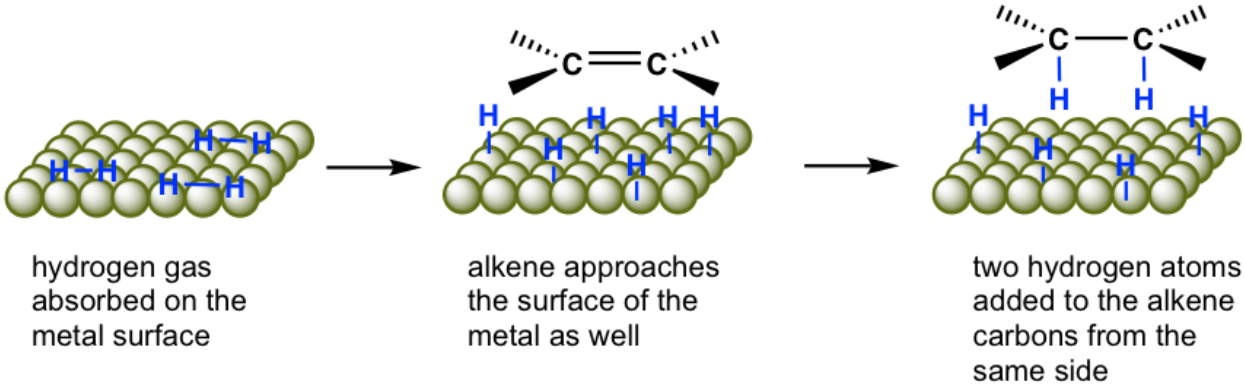
- None

**Stereochemistry:**

- Syn Addition

**Notes:**

- Occurs on metal surface; no carbocation. No rearrangements.




**simplified diagram for catalytic hydrogenation of alkene**

# Vicinal Dihalogenation

**Brief Description:**

The alkene attacks $X_2$, forming a **halonium ion** and releasing $X^-$. The halide ion then attacks from the backside, opening the ring and placing the second halogen on the opposite face.

**Reagent(s):**

1. $Br_2$ or $Cl_2$

**Product(s)**

- Vicinal Dihalide (X & X across DB)

**Intermediate(s):**

- Halonium ion

**Regiochemistry:**

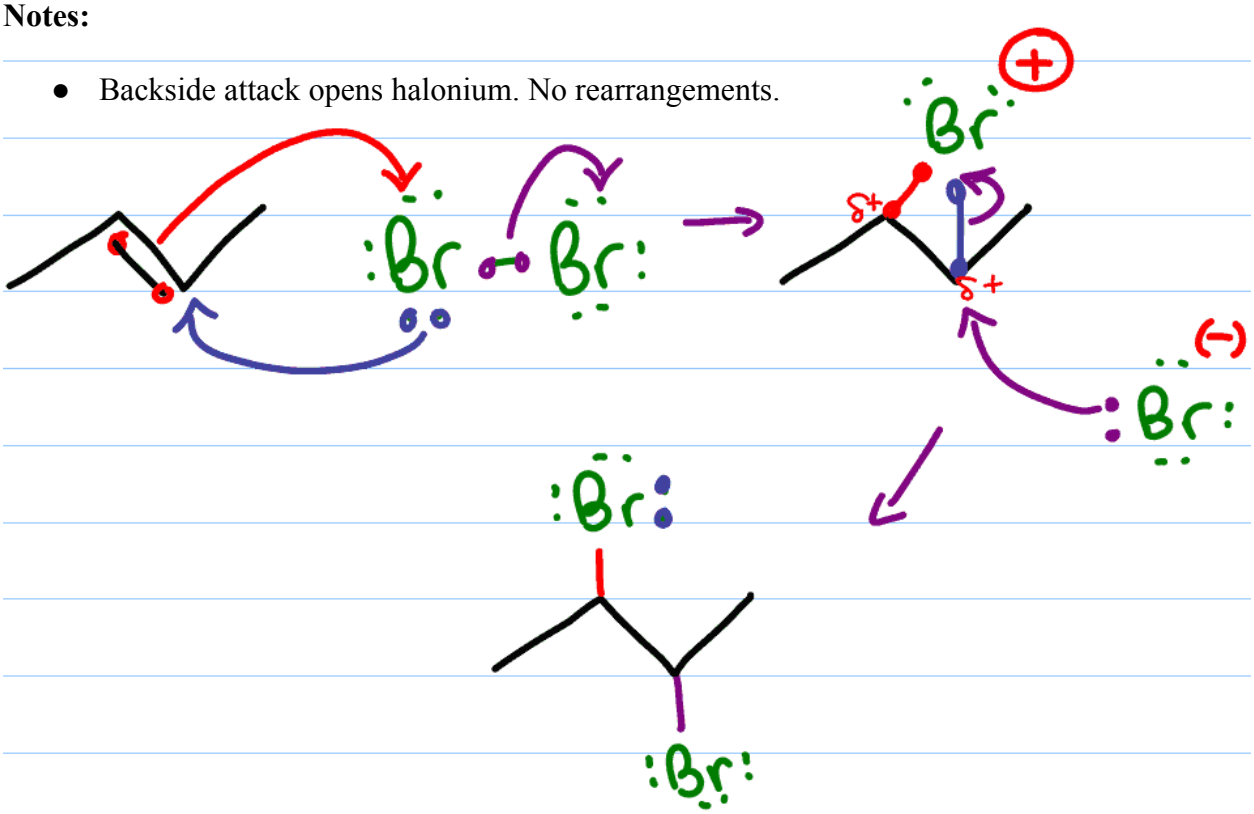
- None

**Stereochemistry:**

- Anti Addition

**Notes:**

- Backside attack opens halonium. No rearrangements.



# Hydrohalogenation

**Brief Description:**

The alkene is protonated by HX, forming the **most stable carbocation**. The halide ion then attacks the planar carbocation from either side, giving a racemic alkyl halide.

**Reagent(s):**

1. HX (HCl, HBr, HI)

**Product(s)**

- Alkyl Halide (H & X added across DB)

**Intermediate(s):**

- Carbocation

**Regiochemistry:**

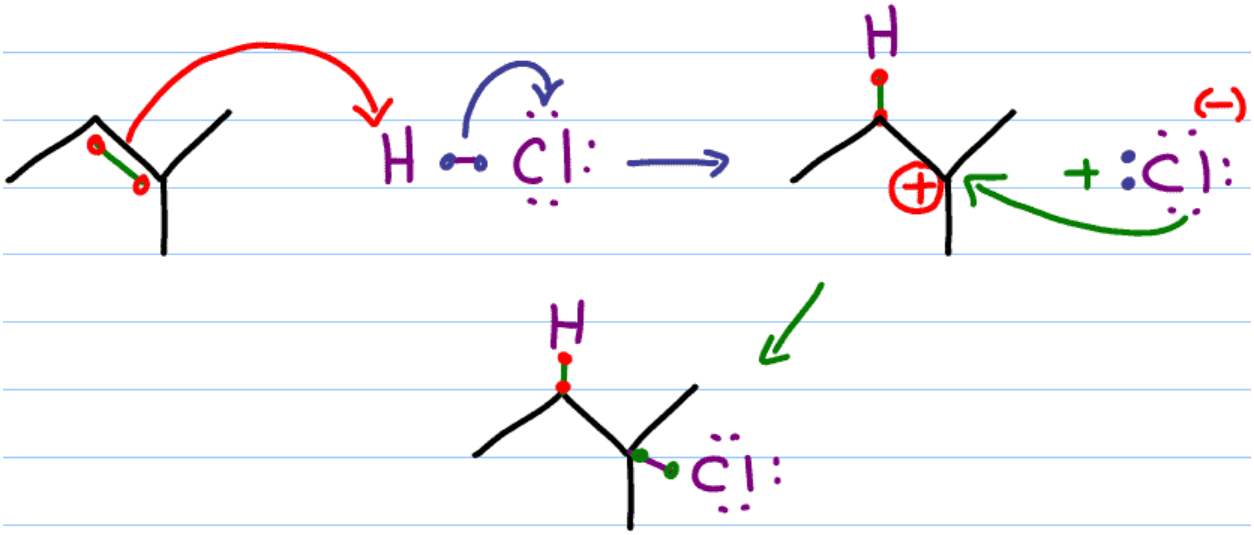
- Markovnikov (X goes to more substituted C)

**Stereochemistry:**

- Racemic (if chiral center forms)

**Notes:**

- Rearrangements possible.



# Hydroboration - Oxidation

**Brief Description:**

$BH_3$ adds across the double bond in a **concerted step**, forming C–B and C–H bonds simultaneously (syn). Boron attaches to the less substituted carbon. In oxidation, $H_2O_2$/NaOH replaces B with OH without changing the carbon framework.

**Reagents:**

1. $BH_3 \cdot THF$
2. $H_2O_2$, NaOH

**Products**:

- Alcohol (OH & H added across DB)

**Intermediate(s)**:

- Concerted transition state (no intermediate)

**Regiochemistry**:

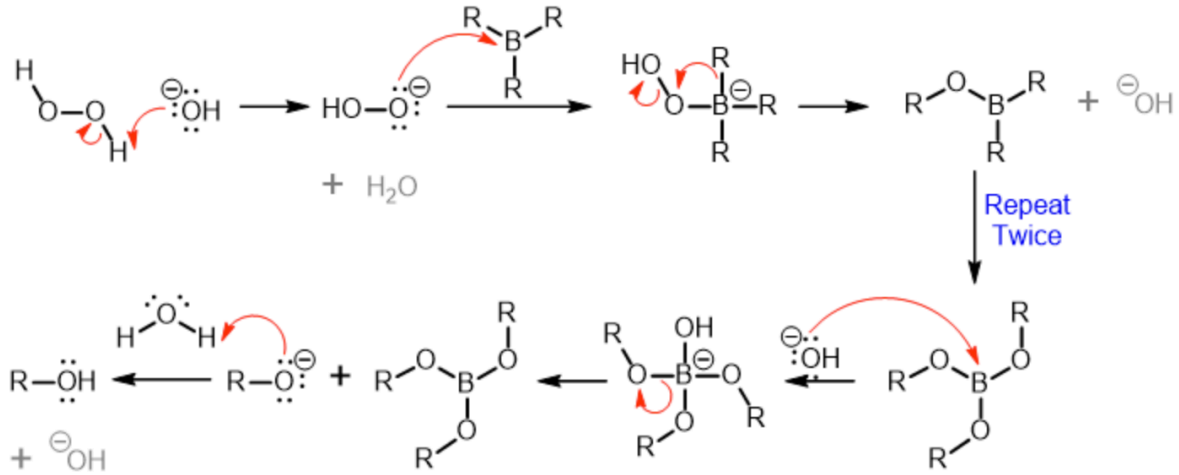
- Anti-Markovnikov (OH to less substituted)

**Stereochemistry**:

- Syn Addition

**Notes**:

- No rearrangements. Complicated mechanism–don't need to know.



# Acid-Catalyzed Hydration of Alkenes

**Brief Description:**

The alkene is protonated to form a **carbocation**. Water attacks the carbocation, forming a protonated alcohol, which is then deprotonated to give the alcohol. Rearrangements can occur before nucleophilic attack.

**Reagents:**

1. $H_2SO_4 + H_2O$ ($H_3O^+$)

**Products**:

- Alcohol (OH & H across DB)

**Intermediate(s)**:

- Carbocation

**Regiochemistry**:

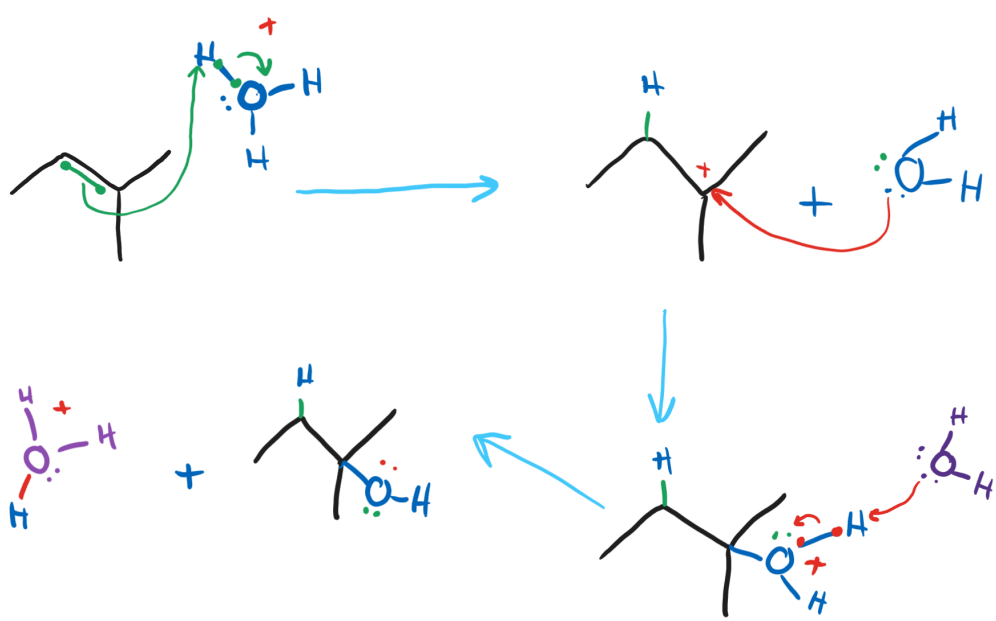
- Markovnikov (OH to more substituted)

**Stereochemistry**:

- Racemic (if chiral)

**Notes**:

- Rearrangements possible



# Haloalcohol (Halohydrin) Formation from Alkenes

**Brief Description:**

The alkene reacts with $X_2$ to form a **halonium ion**. Water (in excess) attacks the more substituted carbon from the backside, opening the ring. Deprotonation gives the halohydrin with OH and X anti.

**Reagents:**

1. $X_2 + H_2O$

**Products**:

- Halohydrin (OH & X added across DB)

**Intermediate(s)**:

- Halonium ion

**Regiochemistry**:

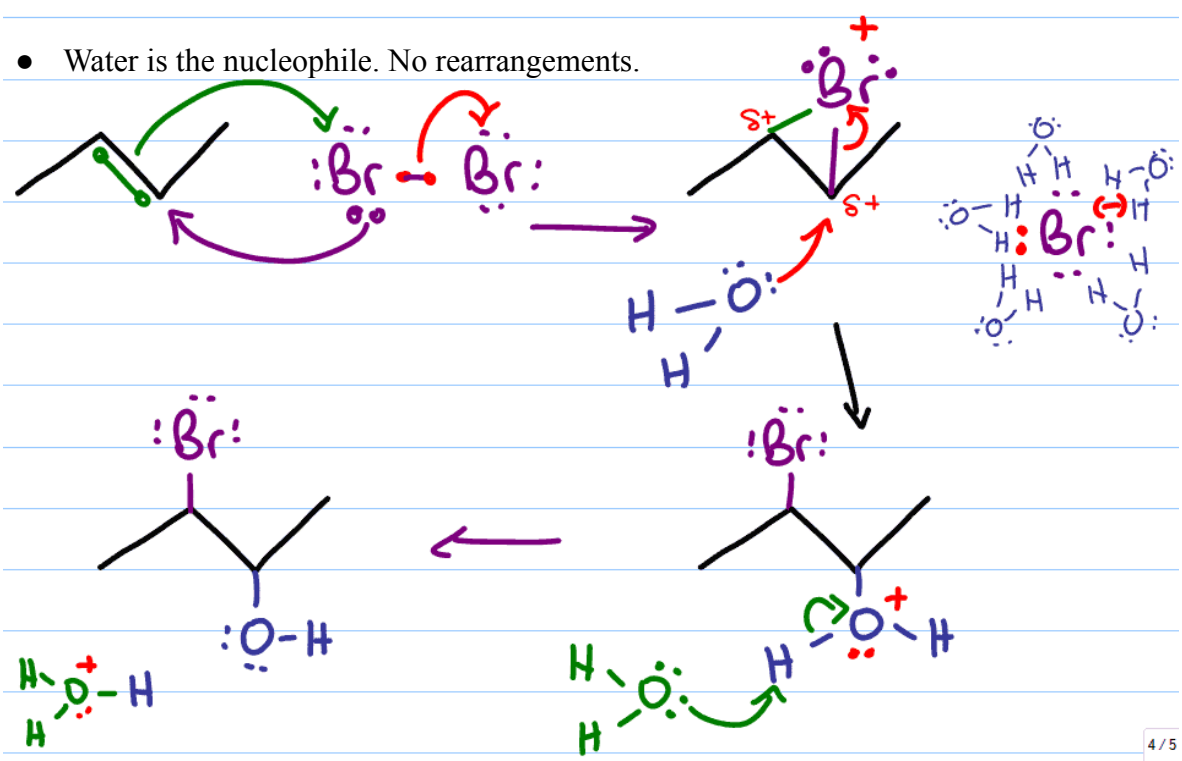
- Markovnikov (OH to more substituted C)

**Stereochemistry**:

- Anti Addition

**Notes**:

- Water is the nucleophile. No rearrangements.



# Haloether Formation from Alkenes

**Brief Description:**

The alkene forms a **halonium ion** with $X_2$. An alcohol (ROH) attacks the more substituted carbon from the backside, opening the ring. Deprotonation gives an ether (–OR) and halogen anti.

**Reagents:**

2. $X_2$ + ROH

**Products**:

- Haloether (OR & X added across DB)

**Intermediate(s)**:

- Halonium ion

**Regiochemistry**:

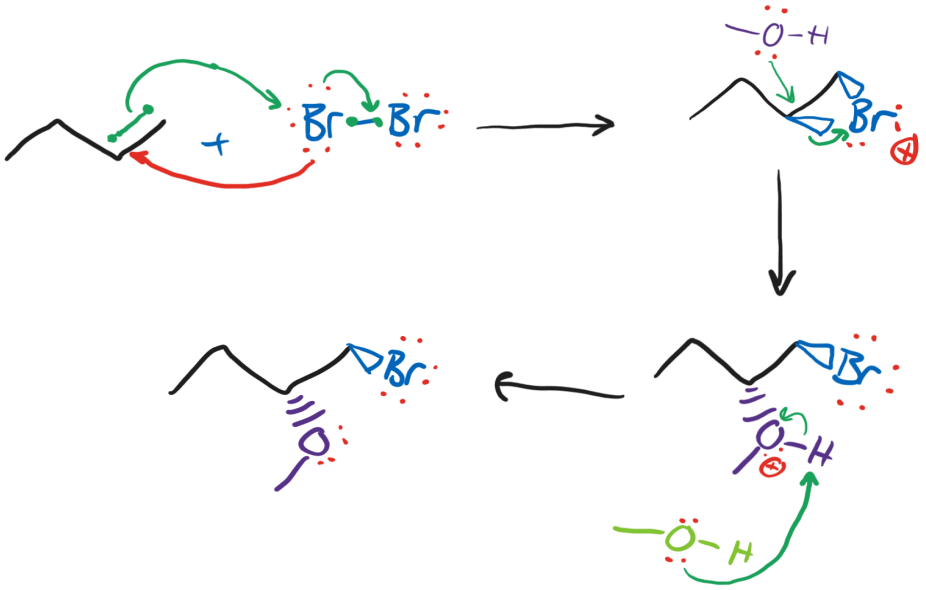
- Markovnikov (OR to more substituted)

**Stereochemistry**:

- Anti Addition

**Notes**:

- ROH acts as the nucleophile instead of water. No rearrangements.



# Epoxide Synthesis

**Brief Description:**

A peroxyacid (mCPBA) transfers an oxygen to the alkene in a **concerted step**. The π bond breaks as a new C–O–C ring forms, giving an epoxide with retention of stereochemistry.

**Reagents:**

1. mCPBA (or peroxyacid) + $CH_2Cl_2$

**Products**:

- Epoxide (3-membered cyclic ether) replaces DB

**Intermediate(s)**:

- Concerted (no discrete intermediate)

**Regiochemistry**:

- None

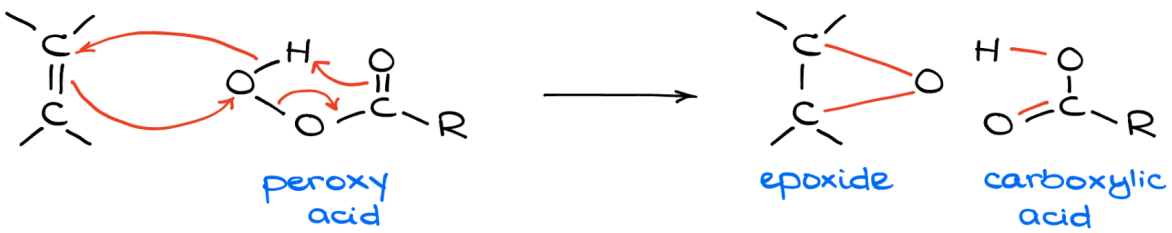
**Stereochemistry**:

- Syn Addition (retains alkene geometry)

**Notes**:

- Stereospecific. No rearrangements.

## Mechanism:



# [Epoxide-Hydrolysis-Mediated] *Vicinal Anti-Hydroxylation* of Alkenes

**Brief Description:**

The epoxide is protonated, making it more electrophilic. Water attacks the more substituted carbon from the backside, opening the ring. Deprotonation yields a vicinal diol with anti stereochemistry.

**Reagents:**

1. mCPBA (or peroxyacid), $CH_2Cl_2$
2. $H_3O^+$

**Products**:

- Vicinal diol (OH & OH added across DB)

**Intermediate(s)**:

- Epoxide

**Regiochemistry**:

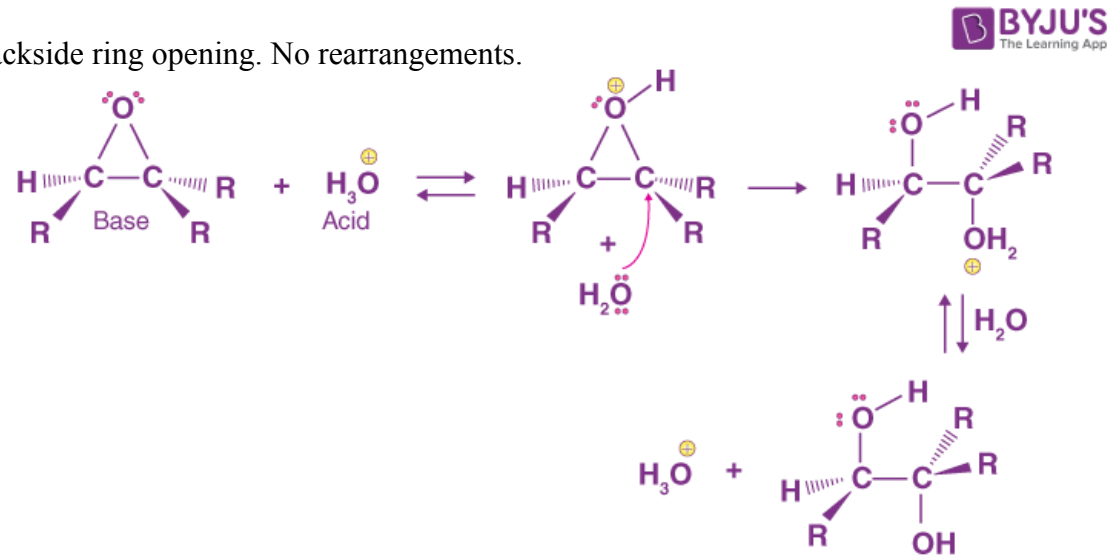
- Water attacks at more substituted carbon (acidic conditions) in an SN2-like fashion, inverting the stereochemistry at the attacked carbon, and opening the epoxide ring.

**Stereochemistry**:

- Anti Addition

**Notes**:

- Backside ring opening. No rearrangements.



# Vicinal Syn-Dihydroxylation

**Brief Description:**

The alkene reacts with $OsO_4$ to form a **cyclic osmate ester**, where both oxygens are added simultaneously to the same face. Reductive workup breaks the Os–O bonds, yielding a syn diol.

**Reagents:**

3. $OsO_4$
4. H2S (aq)

**Products**:

- Vicinal diol (OH & OH added across DB)

**Intermediate(s)**:

- Cyclic osmate ester

**Regiochemistry**:

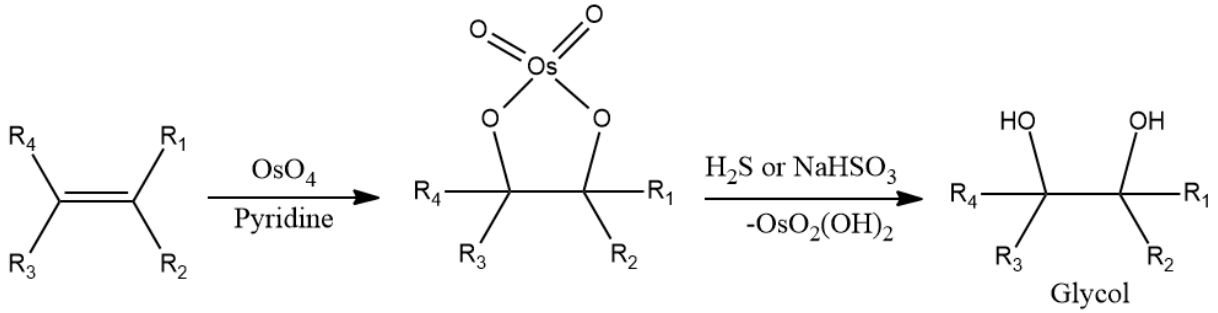
- None

**Stereochemistry**:

- Syn Addition

**Notes**:

- Both OH added to the same face. No rearrangements.



# Ozonolysis

**Brief Description:**

Ozone reacts with the alkene to form an unstable **ozonide intermediate**. Reductive workup (Zn + $H_2O$) breaks the ozonide, splitting the double bond into two carbonyl groups.

**Reagents:**

5. $O_3$
6. Zn + $H_2O$ (or $(CH_3)_2S$) [Reductive workup]

**Products**:

- Aldehydes and/or Ketones

**Intermediate(s)**:

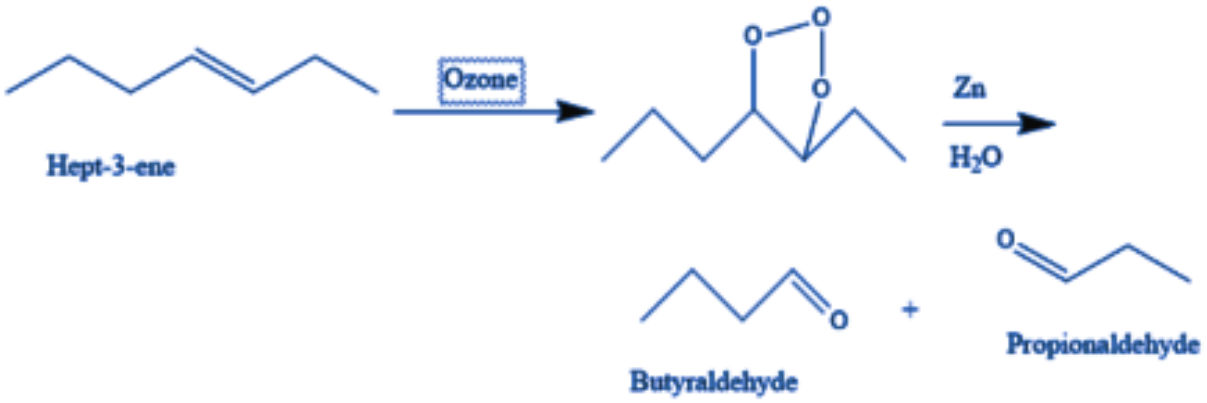
- Ozonide

**Regiochemistry**:

- Cleavage (N/A)

**Stereochemistry**:

- N/A

**Notes**:

- Oxidative workup ($H_2O_2$ instead of Zn + H2O) → carboxylic acids. No rearrangements.